

BHARADWAJ RACHURI

Machine Learning & AI Engineer • MSc AI & Robotics (Commendation) • Published Researcher

Hatfield, UK | 07586 362964 | bharadwaj.r.career@gmail.com | linkedin.com/in/bharadwaj-rachuri | github.com/br23aay | br23aay.github.io

PROFESSIONAL SUMMARY

Machine Learning & AI Engineer with 1 year of industry experience at Swayam Ltd (London), an MSc in Artificial Intelligence and Robotics with Commendation (University of Hertfordshire, 2025), and a peer-reviewed publication in reinforcement learning (IJRES, Impact Factor 7.52). Hands-on experience designing and deploying ML and deep learning models, LLM and RAG pipelines, REST APIs, and robotics simulation environments using PPO and MuJoCo. Skilled in Python, PyTorch, TensorFlow, Azure AI, AWS, and Git. Holder of 49 Microsoft Azure AI certifications. Right to work in the UK — Post-Study Work visa, no sponsorship required. Available immediately.

EXPERIENCE

Machine Learning & AI Engineer | Swayam Ltd, London, UK | May 2025 – June 2026

- Designed, developed, and evaluated machine learning and deep learning models for client-facing projects using Python, TensorFlow, and PyTorch
- Contributed to AI systems incorporating Large Language Models (LLMs), Small Language Models (SLMs), and Vision Models across multiple sectors
- Built and maintained reinforcement learning pipelines using Proximal Policy Optimisation (PPO) and robotics simulation environments in MuJoCo, including sim-to-real transfer workflows
- Developed and deployed RESTful APIs using Flask; integrated with Microsoft Azure AI and AWS cloud platforms
- Maintained CI/CD pipelines and version control using Git and GitHub throughout all client engagements
- Delivered solutions spanning software development, cybersecurity, and AI-assisted development for clients across multiple sectors

EDUCATION

MSc Artificial Intelligence and Robotics — Commendation | University of Hertfordshire, UK | Jan 2024 – Jul 2025

- Published researcher — IJRES Vol. 13, Issue 6, pp. 164–183 | Impact Factor 7.52 | co-authored with Dr Diego Resende Faria
- MSc Project: Trained Shadow Hand (24-DoF) using PPO in MuJoCo — achieved 175° pen rotation, >90% touch sensor accuracy across 3 independent seeds
- Key modules: Machine Learning, Neural Networks, Artificial Life with Robotics, Responsible Technology
- Completed 49 Microsoft Azure AI certifications alongside degree

B. Tech Mechanical Engineering | Guntur Engineering College, JNTUK, India | Aug 2017 – Sep 2021

- Final project: Static Analysis of Spur Gear using FEA (Fusion 360 + ANSYS Workbench)
- Siemens certified: Milling NC Programming and Milling Operating & Machining (2019)

KEY PROJECTS

Dexterous In-Hand Manipulation — Shadow Hand | MSc Project, Published April 2025

- Trained PPO RL agent in custom MuJoCo environment — achieved 175° pen rotation (97.2% of target) and >90% touch sensor accuracy across all 5 fingers
- Reproduced results consistently across 3 independent seeds: 3.118 ± 0.047 rad | $18,295 \pm 487$ peak reward

- Published in IJRES Vol. 13, Issue 6, pp. 164–183 | Impact Factor 7.52 | ISSN: 2320-9364
- Code: github.com/br23aay/shadowhand-dexterity-ppo

Azure AI LLM Workflow Development | 2024–2025

- Built RAG solutions using Azure AI Foundry — retrieval pipelines, context management, output grounding
- Fine-tuned language models and built prompt flow pipelines for structured output generation
- Implemented responsible AI evaluation frameworks and safety guardrails; benchmarked 100+ model outputs

Autonomous Robotic Navigation System — Unity | 2024 | Score: 100/100

- Built sensor-driven autonomous vehicle in Unity using C# — scored 100/100
- Implemented 7-sensor raycast array, lane-keeping logic, and real-time obstacle avoidance
- GitHub: github.com/br23aay/Autonomous-Robotic-Navigation-System-in-Unity

TECHNICAL SKILLS

Languages & Frameworks: Python (proficient), PyTorch, TensorFlow, Keras, Stable-Baselines3, Scikit-learn, Flask, C#, Java, R

LLMs & Generative AI: RAG Pipelines, Fine-tuning (LLaMA, Mistral, GPT), Prompt Engineering, Transformers, Embeddings, RLHF, AI Safety

MLOps & Cloud: Azure AI Foundry, MLflow, Microsoft Fabric, Prompt Flow, AWS, Git, GitHub, CI/CD, REST APIs, Docker

Reinforcement Learning: PPO, SAC, MuJoCo, Gymnasium, Reward Shaping, Sim-to-Real Transfer

Data & ML: NumPy, Pandas, Feature Engineering, Data Pipelines, Model Evaluation, SQL

Robotics: Shadow Dexterous Hand, MuJoCo, Unity Engine, Sensor Integration, Autonomous Systems

Responsible AI: Azure Responsible AI Framework, Safety Guardrails, Bias Detection, Compliance Evaluation

CERTIFICATIONS — 49 MICROSOFT AZURE AI & FABRIC BADGES (AUGUST 2025)

- Azure AI Foundry: RAG solutions, model fine-tuning, prompt flow, SDK applications, responsible AI, model evaluation
- Microsoft Fabric: Data Science, MLflow tracking, batch inference, data wrangling, end-to-end analytics
- Azure AI — NLP & Speech: conversational models, Q&A systems, NER, speech and text translation
- Azure AI — Computer Vision: image/video analysis, object detection, face detection, OCR, image generation
- AI Foundations: AI/ML/GenAI fundamentals, AI agents, information extraction
- In Progress (2026): ML/DL/NLP Bootcamp, TensorFlow 2, PyTorch Bootcamp, Terraform on AWS

PUBLICATION

Rachuri, B. & Faria, D.R. (2025). Reinforcement Learning for Robot Dexterous In-Hand Manipulation of Objects (Shadow Hand). IJRES, Vol. 13, Issue 6, pp. 164–183. ISSN: 2320-9364. Impact Factor: 7.52. ijres.org/papers/Volume-13/Issue-6/1306164183.pdf

ADDITIONAL INFORMATION

- Right to work in the UK — Post-Study Work (PSW) visa, no sponsorship required
- Available for immediate start — open to any UK location including London, Manchester, Birmingham, Bristol, Edinburgh
- Employment reference available from Swayam Ltd, London — confirming 1 year as Machine Learning & AI Engineer